



JSS MAHAVIDYAPEETHA
JSS University, Noida
CIE-1 (AY 2026-27: Odd Semester)



Course: MCA
Subject Code: 25 MCA 331
Max. Marks: 20

Semester: III
Subject Name: Cloud Computing
Date: 17/09/2026 | Time: 1 Hour

COURSE OUTCOMES:

- CO1: Understand cloud computing architecture, service models, deployment models, and core characteristics.
CO2: Explain Inter cloud resources management cloud storage services and their providers Assess security services and standards for cloud computing
CO3: Analyze cloud security challenges, governance models, and security standards including IAM and virtualization security.
CO4: Analyze advanced cloud technologies.
CO5: Design cloud-based solutions addressing real-world challenges including big data processing, security, and performance optimization.

PART A: ANSWER ALL THE QUESTIONS Each question carries 1 mark 4Qx1=04

Q	QUESTIONS	BL	CO
1	A - Differentiate between cloud computing and internet?	1	CO1
	B - A student uses an online application to create, edit and save Documents without installing or maintaining the software on the computer. Identify the cloud service model being used?	2	CO1
	C - A company allows employees to access cloud resources according to their assigned permissions. Identify the cloud security mechanism used to manage these permissions?	1	CO2
	D - A cloud provider serves many customers using shared infrastructure. Identify the cloud environment characteristic represented by this situation?	2	CO2

PART B: ANSWER ALL THE QUESTIONS Each question carries 4 marks 3Qx4=12

Q	QUESTIONS	BL	CO
2	a - A software company wants to develop a web application. It does not want to purchase physical servers or manage operating systems. The developers want to concentrate only on developing and deploying the application. Identify the suitable cloud service model and explain how it satisfies the company's requirement with a suitable example?	3	CO1
	OR		
3	b - An organization has two requirements. For its first application, it needs control over virtual machines, operating systems and storage. For its second application, employees simply need to use ready-made software through the internet. Analyze the suitable cloud service models for both requirements and justify your answer with examples?	4	CO1
	a - Apply the four cloud deployment models—Public, Private, Hybrid, and Community Cloud—to suitable real-world scenarios. Give an example for each model.	4	CO1
4	OR		
	b - Explain any four characteristics of cloud computing with suitable examples.	3	CO1
	a - Explain the role of IAM in securing cloud resources?	3	CO2
4	OR		
	b - Explain any three measures used to ensure security of virtual machines in cloud computing.	3	CO2

PART C: ANSWER ANY 1 QUESTION Each question carries 4 marks 1Qx4=4

Q	QUESTIONS	BL	CO
5	A cloud-based organization experiences a sudden increase in users during an online event. The existing resources are not sufficient to handle the increased workload. Analyze how resource provisioning can be used to address this situation. Explain any two resource provisioning methods with suitable examples?	4	CO2
6	An organization requires additional cloud resources for a short period but does not want to purchase new infrastructure. Analyze how the global exchange of cloud resources can help the organization meet its requirements. Explain its benefits with a suitable example?	4	CO2